In this page, I investigate some typical finite groups.
Specifically, investigate:
Abelian and Abelization
Commutator and Derived subgroup
Upper-Central series and Solvablity and Composition series
Center and Conjugacy classes.
Explicit all subgroup

The general Properties of S_n, D_n, C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy Classes.

Since $|G:G_a| = |C_a|$, find the Centralizer of $(0,(12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$C_G(0,(1,2)) = \{(0,(12)), (0,id), (1,(1,2)), (1,id)\}$.

Therefore, the Conjugacy classes of $(0,(12))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is identity), the first term never change, and since the Conjugate cannot change the cycle structure, the Conjugacy classes of $(0,(12))$ is

$$\{(0,(12)), (0,(13)), (0,(23))\},$$

and similarly, the Conjugacy classes of $(1,(12))$ is

$$\{(1,(12)), (1,(13)), (1,(23))\},$$

" $(0,(123))$ is

$$\{(0,(123)), (0,(132))\}$$

" $(1,(123))$ is

$$\{(1,(123)), (1,(132))\}$$

and $(0,id)$ and $(1,id)$ have Conjugacy classes only containing itself.

$\sqrt{S_3 \times S_3}$, Conjugacy classes.

Let $(1\ 2)\times(1\ 2) \in S_3 \times S_3$, $(1\ 2)$ commute only id or itself, thus

$$C_G((1\ 2)) = \{(id, id), (id, (1\ 2)), ((1\ 2), id), ((1\ 2), (1\ 2))\},$$

So, the conjugacy classes of $(1\ 2)\times(1\ 2)$ has 9 elements

「S_4.

$S_4 =$	$\{ id,$
1,1,2	$(1\,2), (1\,3), (1\,4), (2\,3), (2\,4), (3\,4),$
2,2	$(1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3),$
1,3	$(1\,2\,3), (1\,3\,2), (1\,3\,4), (1\,4\,3), (1\,2\,4), (1\,4\,2), (2\,3\,4), (2\,4\,3),$
4	$(1\,2\,3\,4), (1\,2\,4\,3), (1\,3\,2\,4), (1\,3\,4\,2), (1\,4\,2\,3), (1\,4\,3\,2)\}$
Partition	

A_4. Conjugacy classes.

$A_4 = \{id, (12)(34), (13)(24), (14)(23),$
$(123), (132), (124), (142), (134), (143),$
$(234), (243)\}$

Since the conjugate cannot change the cycle type,
the conjugacy class of each element in A_4 are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(123)(12)(34)(123)^{-1} = (23)(14)$$
$$(124)(12)(34)(124)^{-1} = (24)(31)$$

therefore, the conjugacy classes of $(12)(34)$ equals the set of $(2,2)$ cycles.

$$(132)(123)(132)^{-1} = (312) = (123) \quad (\text{In actually, } (123)^{-1} = (132))$$

$$(124)(123)(124)^{-1} = (243)$$

$$(142)(123)(142)^{-1} = (413) = (134)$$
$$(134)(123)(134)^{-1} = (324) = (243)$$
$$(143)(123)(143)^{-1} = (421) = (142)$$
$$(234)(123)(234)^{-1} = (134)$$
$$(243)(123)(243)^{-1} = (142)$$
$$(12)(34)(123)((12)(34))^{-1} = (214)$$
$$(13)(24)(123)(13)(24) = (341) = (134)$$
$$(14)(23)(123)(14)(23) = (432) = (243)$$

therefore, the conjugacy classes of (123) has just four elements.
Generally, since $|G:G_a| = C_a$ and $G_a = C_G(a)$ in the conjugate action, therefore number of elements of the orbit is $|G|/|C_G(a)|$.
Since $C_{A_4}((123)) = \{id, (123), (132)\}$,

the orbit of (123) has $12/3 = 4$ elements.

A_5. The Smallest group of non-abelian Simple group.

「Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi : G \to S_4$;

$(0,0) \mapsto id$, $(1,0) \mapsto (1\,2)(3\,4)$, $(0,1) \mapsto (1\,3)(2\,4)$, $(1,1) \mapsto (1\,4)(2\,3)$

is Embedding